Problem name: **Multiples(1)**

Summary of problem:
Two integer number will be given. If larger number is divisible by small number then print "Multiples". Otherwise print "No multiples".

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define mod0(x,y) x%y==0
#define newl cout<<endl
int main()
{
   int a,b;
   cin>>a>>b;
   mod0(a,b)||mod0(b,a)?cout<<"Multiples":cout<<"No Multiples";
}
```
Output

Input:	**Output:**
9 3. 12 5	Multiples No Multiples

Screenshot of Accepted Bar

217203682	Practice: Arsenic33	219158J - 20	GNU C++20 (64)	Accepted	15 ms	8 KB	2023-08-05 08:52:45	2023-08-05 08:52:45

Comments of mine:
It was conditional statement using problem. I had to check if the largest number is divisible by small number or not. Here I used Ternary operator to solve this problem. The time complexity is O(1) and Space Complexity is O(1).

Problem name: **Sort Numbers(2)**

Summary of problem:
Three integer numbers will be given. We have to sort it in increasing order and print them and then print the actual order of the numbers.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define __lcm(x,y) (x*y)/__gcd(x,y)
#define A_sort(x) sort(x.begin(),x.end())
#define newl cout<<endl
int main()
{
   vector<int>A(3),B;
   cin>>A[0]>>A[1]>>A[2];
   B=A;
   A_sort(A);
   for(int i:A)
   {
      cout<<i;newl;
   }
   newl;
   for(int i:B)
```

```
    {
        cout<<i;newl;
    }
}
```
Output

Input: 3 -2 1	**Output:** -2 1 3 3 -2 1

Screenshot of Accepted Bar:

217203307	Practice: Arsenic33	219158T - 12	GNU C++20 (64)	Accepted	0 ms	8 KB	2023-08-05 08:49:10	2023-08-05 08:49:10

Comments of mine:
Here I have used vector to store the value of 3 numbers and sort them in increasing order by using another vector and print them and also print the actual order of the numbers. Time complexity is O(1) and Space Complexity is O(1).

Problem name: **Multiplication Table(3)**

Summary of problem:
There will be given a number and print the 1 to 12^{th} multiplication of that number.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
    int a;
    cin>>a;
    for(int i=1;i<=12;i++)
    {
        cout<<a<<" * "<<i<<" = "<<a*i<<endl;
    }
}
```
Output:

Input: 1	**Output:** 1 * 1 = 1 1 * 2 = 2 1 * 3 = 3 1 * 4 = 4 1 * 5 = 5 1 * 6 = 6 1 * 7 = 7 1 * 8 = 8 1 * 9 = 9 1 * 10 = 10 1 * 11 = 11 1 * 12 = 12

Screenshot of Accepted Bar:

217204219	Practice: Arsenic33	219432F - 14	GNU C++20 (64)	Accepted	15 ms	12 KB	2023-08-05 08:57:47	2023-08-05 08:57:47

Comments of mine
It was the problem of loop and done some multiplication to get the table. Time complexity is O(1) and Space Complexity is O(1).

Problem name: **Even, Odd, Positive and Negative(4)**

Summary of problem:
There will be given N numbers and print total even, odd, positive and negative number.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
  int n;cin>>n;
  int E=0,O=0,P=0,N=0;
  for(int i=1;i<=n;i++)
  {
    int a;cin>>a;
    if(a%2==0)
    {
      E++;
    }
    else
    {
      O++;
    }
    if(a<0)
    {
      N++;
    }
    else if(a>0)
    {
      P++;
    }
  }
  cout<<"Even: "<<E<<endl;
  cout<<"Odd: "<<O<<endl;
  cout<<"Positive: "<<P<<endl;
  cout<<"Negative: "<<N;
}
```
Output:

Input:	**Output:**
5 -5 0 -3 -4 12	Even: 3 Odd: 2 Positive: 1 Negative: 3

Screenshot of Accepted Bar:

208524686	Practice: Arsenic33	219432C - 19	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-06-04 22:52:53	2023-06-04 22:52:53

Comments of mine:
A number could be even or odd and positive or negative. But here 0 is not positive or negative number , so it was not counted. Time complexity is O(N) and Space Complexity is O(1).

Problem name: **Three Numbers(5)**

Summary of problem:
There will be given 2 numbers (K & S). Print how many different values of X,Y,Z such that (0<=X,Y,Z<=k) and X+Y+Z=S.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
   int a,b;
   cin>>a>>b;
   ll x=0;
   for(int i=0;i<=a;i++)
   {
      for(int j=0;j<=a;j++)
      {
         if((b-i-j)>=0&&(b-i-j)<=a)
         {
            x++;
         }
      }
   }
   cout<<x;
}
```
Output:

Input:	**Output:**
9 4	15

Screenshot of Accepted Bar:

217206394	Practice: Arsenic33	219432Z - 9	GNU C++20 (64)	Accepted	31 ms	12 KB	2023-08-05 09:16:13	2023-08-05 09:16:13

Comments of mine:
It was a conditional and looping problem. I have to use nested loop for implementing this problem. Time complexity is O(n^2) and Space Complexity is O(1).

Problem name: **Beautiful Matrix(6)**

Summary of problem:
There will be given 5*5 matrix and there will only 1 in one cell and other cell are 0. Find out the minimum steps to place the 1 in the cell (3,3) by swapping with each other.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
   int r[6][6],x,y;
   for(int i=1;i<=5;i++)
```

```
  {
    for(int j=1;j<=5;j++)
    {
      cin>>r[i][j];
      if(r[i][j]==1)
      {
        x=i;
        y=j;
      }
    }
  }
  int a=x-3,b=y-3;
  if(a<0)
  {
    a*=-1;
  }
  if(b<0)
  {
    b*=-1;
  }
  int d=a+b;
  cout<<d;
}
```

Output:

Input:	**Output:**
0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0	3

Screenshot of Accepted Bar:

217207411	Practice: Arsenic33	263A - 8	GNU C++20 (64)	Accepted	0 ms	8 KB	2023-08-05 09:24:43	2023-08-05 09:24:43

Comments of mine:
Here I have used Manhathan distance to calculate the steps. Time complexity is O(1) and space complexity is O(1).

Problem name: **One Prime(7)**

Summary of problem:
There will be given an integer number and have to be determined whether it is prime or composite number.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
  int a;
  cin>>a;
  for(int i=2;i<a;i++)
  {
    if(a%i==0)
    {
      cout<<"NO";
      return 0;
```

```
        }
    }
    cout<<"YES";
}
```

Output:

Input: 7 15	**Output:** YES NO

Screenshot of Accepted Bar:

208525406	Practice: Arsenic33	219432H - 27	GNU C++20 (64)	Accepted	15 ms	0 KB	2023-06-04 23:04:25	2023-06-04 23:04:25

Comments of mine:
If the number is divided by only 1 and that number it is called a prime number. So I have tested that number if it is divisible by 2 to (n-1). Time complexity O(n) and Space complexity is O(1).

Problem name: **Increasing Array(8)**

Summary of problem:
There will be given n integers and. I have to determine the steps to make the numbers will in non-decreasing order.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
    int n;
    cin>>n;
    vector<int>A(n);
    for(int i=0;i<n;i++)
    {
        cin>>A[i];
    }
    ll x=0;
    for(int i=1;i<n;i++)
    {
        if(A[i]<A[i-1])
        {
            x+=(A[i-1]-A[i]);
            A[i]=A[i-1];
        }
    }
    cout<<x;
}
```

Output:

Input: 5 3 2 5 1 7	**Output:** 5

Screenshot of Accepted Bar:

Task:	[Increasing Array]
Sender:	Arsenic
Submission time:	2023-08-05 09:52:05 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:
Here I have just checked if a number is lower than the previous then increased that number to equal with previous one. Time complexity and memory complexity is O(n)

Problem name: **Missing Number(9)**

Summary of problem:
there will be given (n-1) numbers and I have to find the missing number as they are consecutive.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
   ll n;cin>>n;
   set<int>s;
   for(int i=1;i<=n;i++)
   {
      s.insert(i);
   }
   for(int i=1;i<=n-1;i++)
   {
      int a;cin>>a;
      s.erase(a);
   }
   for(auto i:s)
   {
      cout<<i;
   }
}
```
Output:

Input: 5 2 3 1 5	**Output:** 4

Screenshot of Accepted Bar:

Task:	[Missing Number]
Sender:	Arsenic
Submission time:	2023-03-31 15:01:43 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:
Here I have used a data structure set where unique numbers are stored in consecutively. I have erased the non -missing numbers and found the missing. Time complexity and Space Complexity is O(n).

Problem name: **Magic Elements(10)**

Summary of problem:
There will be given n integers and a number k. I have to determine whether a number is large than the summation of other numbers by adding k.

Code:

```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
   int t;
   cin>>t;
   while(t--)
   {
      int n,k;
      cin>>n>>k;
      vector<int>A(n);
      ll sum=0;
      for(int i=0;i<n;i++)
      {
         cin>>A[i];
         sum+=A[i];
      }
      int x=0;
      for(int i=0;i<n;i++)
      {
         if((sum-A[i])<(A[i]+k))
         {
            x++;
         }
      }
      cout<<x;
      newl;
   }
}
```

Output:

Input: 2 4 4 2 1 6 7 4 2 2 1 5 4	**Output:** 1 0

Screenshot of Accepted Bar:

Comments of mine:
To lessen the time complexity of the I have done the summation of that number and done the next step. Time complexity and Space Complexity is O(n).

Problem name: **Playlist(11)**

Summary of problem:
There will be given n integers and I have to find out the longest subarray of unique numbers.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main() // by using sliding window technique......................
{
   int n;
   cin>>n;
   int mx=-100;
   vector<int>A(n);
   for(int i=0;i<n;i++)
   {
      cin>>A[i];
   }
   int l=0,r=0;
   map<int,pair<int,int>>mp;
   while(r<n)
   {
      if(r==0)
      {
         mp[A[r]].first=1;
         mp[A[r]].second=r;
         mx=max(mx,r-l+1);
         r++;
      }
      else if(mp[A[r]].first==0)
      {
         mp[A[r]].first=1;
         mp[A[r]].second=r;
         mx=max(mx,r-l+1);
         r++;
      }
      else if(mp[A[r]].first==1)
      {
         mp[A[l]].first=0;
         l++;
      }
   }
   cout<<mx<<endl;
}
```
Output:

Input:	**Output:**
8 1 2 1 3 2 7 4 2	5

Screenshot of Accepted Bar:

Task:	Playlist
Sender:	Arsenic
Submission time:	2023-08-18 20:57:27 +0300
Language:	C++20
Status:	READY
Result:	ACCEPTED

Comments of mine:
Here I have used sliding window technique to determine the longest subarray of unique number. Time complexity and space Complexity is O(n).

Problem name: **Word Capitalization(12)**

Summary of problem:
There will be given a string and print the string with first character in uppercase letters.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
   str r;
   cin>>r;
   if(r[0]>='a'&&r[0]<='z')
   {
      r[0]-=32;
      cout<<r;
   }
   else
   {
      cout<<r;
   }
}
```
Output:

Input: ApPLe konjac	**Output:** ApPLe Konjac

Screenshot of Accepted Bar:

219486357	Practice: Arsenic33	281A - 22	GNU C++20 (64)	Accepted	30 ms	0 KB	2023-08-18 21:09:03	2023-08-18 21:09:03

Comments of mine:
It was a string based problem. Time complexity is O(1) and Space Complexity is O(string size).

Problem name: **Way Too Long Words(13)**

Summary of problem:
There will be given a string and have to determine if the size is greater than 10. if it then print first character ,size-2, last character. Otherwise print that string.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
int main()
{
   int t;
   cin>>t;
   while(t--)
   {
      str r;
      cin>>r;
      int n=r.size();
      n>10?cout<<r[0]<<(n-2)<<r[n-1]:cout<<r;
      newl;
```

```
    }
}
```
Output:

Input: 4 word localization internationalization pneumonoultramicroscopicsilicovolcanoconiosis	**Output:** word l10n i18n p43s

Screenshot of Accepted Bar:

219486770	Practice: Arsenic33	71A - 14	GNU C++20 (64)	Accepted	0 ms	0 KB	2023-08-18 21:12:55	2023-08-18 21:12:55

Comments of mine
It was also a string based problem. Time complexity is O(1) and Space Complexity is O(string size).

Problem name: **Float or Int(14)**

Summary of problem:
There will be given a number and I have to print integer part and decimal part if it is a float number.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define newl cout<<endl
int main()
{
    float x,y;
    cin>>x;
    y=x-int(x);
    y>0?cout<<"float "<<int(x)<<" "<<y:cout<<"int "<<int(x);
}
```
Output:

Input: 234.000 534.958	**Output:** int 234 float 534 0.958

Screenshot of Accepted Bar:

219487350	Practice: Arsenic33	219158U - 15	GNU C++20 (64)	Accepted	15 ms	12 KB	2023-08-18 21:17:55	2023-08-18 21:17:55

Comments of mine:
Here if it is a integer then firstly I have printed int . Otherwise float. Time complexity and Space Complexity is O(1).

Problem name: **Petya and String(15)**

Summary of problem:
There will be given 2 strings and I have to compare the string in lexicographically. If first's string letter is greater than the second one then I have to print 1 if it is reverse then -1. Otherwise 0.

Code:

```
#include<bits/stdc++.h>
#include<string.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
void upp(str &r,str &s)
{
   int n=r.size();
   for(int i=0;i<n;i++)
   {
      if(r[i]>='A'&&r[i]<='Z')
      {
         r[i]+=32;
      }
      if(s[i]>='A'&&s[i]<='Z')
      {
         s[i]+=32;
      }
   }
}
int main()
{
   str r,s;
   cin>>r>>s;
   upp(r,s);
   bool flag=true;
   for(int i=0;i<r.size();i++)
   {
      if(r[i]>s[i])
      {
         flag=false;
         cout<<1;
         break;
      }
      else if(r[i]<s[i])
      {
         flag=false;
         cout<<-1;
         break;
      }
   }
   if(flag)
   {
      cout<<0;
   }
}
```

Output:

Input:	**Output:**
aaaa aaaA abs Abz abcdefg AbCdEfF	0 -1 1

Screenshot of Accepted Bar:

219488437	Practice: Arsenic33	112A - 11	GNU C++20 (64)	Accepted	30 ms	8 KB	2023-08-18 21:27:25	2023-08-18 21:27:25

Comments of mine:
First of all , I have made all the character in lower case and then compared them by running a loop. Time complexity and Space complexity are O(string size).

Problem name: **Find a word(16)**

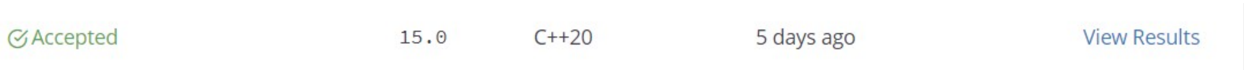
Summary of problem:
There will be given n strings and some condition to think it as a word and find out the total count of a word.

Code:
```
#include<bits/stdc++.h>
using namespace std;
#define ll long long int
#define str string
#define newl cout<<endl
int main()
{
    map<str,int>mp;
    int t;
    cin>>t;
    getchar();
    while(t--)
    {
        str r,s="";
        getline(cin,r);
        for(int i=0;i<r.size();i++)
        {
            if(r[i]>=65&&r[i]<=90||r[i]>=97&&r[i]<=122||r[i]=='_')
            {
                s+=r[i];
            }
            else
            {
                if(s.size()>0)
                {
                    mp[s]++;
                    s="";
                }
            }
        }
        if(s.size()>0)
        {
            mp[s]++;
        }
    }
    int x;cin>>x;
    while (x--)
    {
        str k;
        cin>>k;
        cout<<mp[k];newl;
    }
}
```

Output:

Input: 1 foo bar (foo) bar foo-bar foo_bar foo'bar bar-foo bar, foo. 1 foo	**Output:** 6

Screenshot of Accepted Bar:



Comments of mine:
Here I used map to count the words and finally printed the total number of a word. Time complexity is (max(string size , x)) and space complexity is O(t*string size).